

Huey's Home Lab Build Documentation

Phase 4 — Splunk SIEM Installation & Real-Time Attack Detection

Session Date: June 4, 2026

1. Phase 4 Overview

This document covers Phase 4 of the cybersecurity home lab build. Building on the Active Directory attacks performed in Phase 3, this phase focused on the defensive side — installing Splunk Enterprise as a SIEM, ingesting Windows Security logs, writing SPL detection queries, and creating a real-time alert that successfully detected the CrackMapExec attack from Kali Linux. This phase completes the full attack and detection pipeline.

2. Splunk Installation

2.1 Installation Details

Setting	Value
Product	Splunk Enterprise
Version	Latest (Free 60-day trial)
Installed On	Windows Server 2025 (192.168.99.10)
Web Interface	http://localhost:8000
Admin Username	admin
Admin Password	Adman123#
License	60-day free trial — full features

2.2 Installation Method

Since Windows Server 2025 has no internet access on the Inside network, Splunk was downloaded on the host laptop and transferred to the VM using VMware Shared Folders:

- Downloaded Splunk Enterprise .msi installer on host laptop from splunk.com
- Enabled VMware Shared Folders in VM Settings > Options > Shared Folders
- Set shared folder to host laptop Downloads directory
- Accessed shared folder inside Windows Server via \\vmware-host\Shared Folders
- Double-clicked the .msi installer and completed installation
- Splunk automatically launched in browser at http://localhost:8000 after installation

3. Data Input Configuration

Configured Splunk to ingest Windows Security Event Logs in real time:

- Settings > Add Data > Monitor > Local Event Logs
- Selected the Security log source
- Left all defaults and clicked Submit
- Splunk immediately began ingesting all Windows Security events

Result: Within seconds of configuration Splunk had ingested thousands of Security log events including all historical events and began monitoring in real time.

4. SPL Detection Queries

4.1 Basic Event ID 4624 Search

First search to find all successful logon events:

```
index=main source="WinEventLog:Security" EventCode=4624
```

Result: 5,391 events returned covering the last 7 days of Windows Security log activity.

4.2 Kali-Specific Detection Query

Filtered to show only logons originating from the Kali attack machine:

```
index=main source="WinEventLog:Security" EventCode=4624 Workstation_Name=kali
```

Result: 1,801 events — all network logons where the source workstation was named KALI. The timeline view showed two distinct spikes corresponding exactly to our Phase 3 attack sessions.

4.3 Attack Summary Statistics Query

Grouped results by account and workstation to produce a SOC dashboard view:

```
index=main source="WinEventLog:Security" EventCode=4624 Workstation_Name=kali  
| stats count by Account_Name, Workstation_Name
```

Results:

Account_Name	Workstation_Name	Count	Significance
(unknown)	KALI	1,800	Background system authentications
ANONYMOUS LOGON	KALI	14	Anonymous enum4linux scan from Phase 3
ksayian	KALI	1,786	CrackMapExec authenticated attack — primary IOC
ksayian	kali	1	Single additional authenticated connection

5. Real-Time Alert Configuration

5.1 Alert Settings

Setting	Value	Reason
Title	Suspicious Authentication from Kali	Descriptive name for SOC triage
Description	Detects successful logons from a workstation named KALI which may indicate CrackMapExec or similar SMB attack tools	Context for analyst investigating the alert
Alert Type	Real-time	Fires immediately when attack occurs, not on a schedule
Trigger Condition	Number of Results > 0	Any match is suspicious enough to alert
Trigger	For each result	Alert fires for every individual event detected
Throttle	Enabled — suppress by Workstation_Name for 60 seconds	Prevents alert fatigue from thousands of duplicate alerts
Severity	High	Authentication from a Linux attack machine warrants immediate investigation
Action	Add to Triggered Alerts	Appears in SOC alert queue for analyst review

5.2 Alert Trigger History

After saving the alert, CrackMapExec was run again from Kali to test detection:

- CrackMapExec SMB scan executed from 192.168.99.20
- Windows Server logged Event ID 4624 with Workstation_Name=kali
- Splunk ingested the event in real time
- Alert fired at: 2026-06-04 22:27:54 Eastern Daylight Time
- Trigger History showed 1 confirmed trigger
- View Results confirmed: Account_Name=(-), Workstation_Name=kali, count=1

The complete attack-to-detection pipeline was confirmed working end to end.

6. Complete Attack Detection Pipeline

Step	Component	Action	Result
1	Kali Linux	CrackMapExec SMB scan executed	Attack launched from 192.168.99.20
2	Windows Server	Event ID 4624 logged	Forensic evidence captured in Security log
3	Splunk	Log ingested in real time	Event searchable within seconds
4	Splunk Alert	Detection rule matched	Alert fired at 22:27:54 EDT
5	SOC Analyst	Alert appears in Triggered Alerts	Investigation can begin immediately

7. Key SPL Concepts Learned

SPL Command	Purpose	Example
index=	Specifies which Splunk index to search	index=main
source=	Filters by data source	source="WinEventLog:Security"
EventCode=	Filters by Windows Event ID	EventCode=4624
Workstation_Name=	Filters by source workstation	Workstation_Name=kali
stats count by	Groups and counts results by field	stats count by Account_Name

8. Indicators of Compromise (IOCs) Identified

The following IOCs were identified through Splunk analysis that would indicate an active attack:

IOC	Value	Significance
Source Workstation	KALI	Linux machine named kali authenticating to Windows domain
Source IP	192.168.99.20	Kali Linux machine IP address
Event ID	4624	Successful network logon (Logon Type 3)
Authentication Package	NTLM V2	CrackMapExec uses NTLM by default
Logon Process	NtLmSsp	SMB-based authentication fingerprint
Account	ksayian	Domain user account used in attack

IOC	Value	Significance
Frequency	1,786 events in minutes	Impossible for legitimate user activity

9. Current Lab Status

Component	Status	Details
VMware Workstation Pro	Installed	Version 26H1
GNS3 + GNS3 VM	Configured	Version 2.2.59, Project: Huey's Starship
Huey's Enterprise (Kali)	Domain Joined	192.168.99.20 / hueylab.local
Windows Server 2025	Domain Controller + SIEM	192.168.99.10 / WIN-430KUCVRBB4.hueylab.local
Active Directory	Configured	Domain: hueylab.local, OU: The Sayian Race
Splunk Enterprise	Installed & Running	http://localhost:8000, ingesting Security logs
Detection Alert	Active	Suspicious Authentication from Kali — Real-time, High severity
Attack Pipeline	Verified	Full attack-to-detection cycle confirmed working

10. Next Steps for Phase 5

- Build a Splunk dashboard with multiple detection panels
- Create additional alerts for Event ID 4625 (failed logons) to detect brute force attempts
- Create alert for Event ID 4648 (logon with explicit credentials) to detect lateral movement
- Install BloodHound on Kali for Active Directory attack path visualization
- Practice Pass-the-Hash attacks using captured NTLM hashes
- Remediate the vulnerabilities identified in Phase 3 (password policy, account lockout)
- Add a Windows 10 workstation VM as a third lab machine for lateral movement practice
- Set up Sysmon on Windows Server for enhanced endpoint visibility in Splunk